Participant Profile Form

MSCA Staff Exchanges 2027 (MSCA-SE 2027) — consortium-building process

Unidad de Excelencia IOBA | University of Valladolid

Purpose:
This form is intended to collect structured information from interested participants for the preparation of an MSCA Staff Exchanges consortium. It is designed to support both (i) later completion of participant-related information requested in the EU Funding & Tenders Portal, and (ii) the creation of an internal profile database to help identify complementarities and facilitate contacts across the emerging consortium.

Please complete one form per:

- non-academic organisation (SMEs, NGOs , patient organisations)
- participating academic team (one form per PI, research-group, department).

This prospective **MSCA Staff Exchanges 2027 consortium** aims to build an international, interdisciplinary and inter-sectoral network in eye and vision research, centred on translational collaboration and staff mobility. Starting from Unidad de Excelencia IOBA's strengths in advanced therapies, biomarkers, inflammation, neuroprotection and health technology assessment, the network remains open to additional interdisciplinary collaborations, including areas such as ocular surface, retina, inherited and age-related eye disease, neuro-vision, optometry, drug delivery, AI-enabled diagnosis and patient-oriented validation, while remaining fully open to any other complementary expertise that could enrich the consortium and strengthen its patient-focused impact.

Important:
- please use this form to capture your general interests, expertise, and rough capacity to send and/or host staff.
- If you are not sure how to answer any field, you may leave it blank and highlight it.

1. Organisation and administrative profile

Organisation legal name	International Iberian Nanotechnology Laboratory
Short name / acronym used in projects	INL
Participant Identification Code (PIC), if available	If no PIC is available yet, please indicate 'not yet available'.
Website(s)	Inl.int
Country	Portugal
Organisation type	☐ Academic x Non-academic ☐ Hospital / clinic ☐ SME ☐ Large company ☐ Research infrastructure ☐ NGO / foundation ☐ Public body x Other
Main address	Avenida Mestre José Veiga s/n, 4715-330 | Braga, Portugal
Department(s), institute(s), research group(s) or unit(s) that would be involved	Pastrana Research Group Conference, Communication and marketing Unit
Additional institutional resources relevant to collaboration	https://inl.int/services/research-core-facilities/

Main contact person for this profile	Lorenzo Pastrana, Professor, Research Group Leader
Main contact email	Lorenzo.pastrana@inl.int
Other contact person(s), if relevant	Marian Debs

2. Scientific / innovation profile and relevance for an IOBA-led consortium

Please describe your profile at the level of the participating team / unit most relevant to the prospective consortium.

Indicative thematic relevance (tick all that apply)	
☐ Ocular surface	☐ Retina
☐ Inherited retinal dystrophies (IRDs)	☐ Age-related ocular disease
☐ Neuro-vision / integrated vision / neuro-ophthalmology	☐ Optometry / visual behaviour / rehabilitation
☐ Advanced therapies / regenerative medicine	☐ Gene therapy / cell therapy / tissue engineering
☐ Inflammation / biomarkers / neuroprotection	x Drug delivery / formulations / nanomedicine
☐ Drug discovery / screening / repurposing	☐ Ocular vascular disease
☐ Infection / trauma-related ocular disease	☐ Ocular pain / quality of life / patient-reported outcomes
☐ Health technology assessment	☐ AI-assisted diagnosis / software / digital tools
☐ Clinical trials / translational validation	☐ Other (please specify):

Free Keywords for internal searchable database (please provide up to 10)	Biopolymers, responsive, functionalization, micro and nanoencapsulation, morphological and functional characterization, controlled release
Geographies of interest for exchanges (optional)	No limits

Overview of your research / innovation focus
Pastrana's Research Group pioneers innovative nanotechnology-based solutions to address global challenges and build a healthier, sustainable, fair, and digital food system. This group covers TRL 2 to 7 by developing new active and smart functionalities using biomaterials based on their rheological/textural properties to engineering controlled and smart drug delivery systems by using a large variety of micro and nanoencapsulation technologies including nanospraydryer and ultrasonic atomization among others. In addition, the morphological (TEM, SEM CryoTEM, XPS, SAXS) and functional characterization of the micro and nanoparticles is also a research line. The group develop as well organ on a chip microfluidic devices for assessing digestibility of nanomaterials

Why your team could be relevant to a translational eye-health and vision consortium
Pastrana's Research Group at INL could bring strong added value to a consortium focused on visual health and translational ophthalmology through its expertise in nanotechnology, biomaterials, controlled delivery systems and advanced micro/nanoencapsulation technologies. Although the group's core application area is the development of healthier, sustainable and digital food systems, many of its enabling technologies are highly transferable to ophthalmology, particularly where the field requires new strategies for protecting, stabilising and delivering sensitive bioactive compounds. A key contribution would be the group's experience in engineering biomaterial-based systems with tailored rheological, textural and functional properties. This expertise could support the design of advanced formulations, soft materials or biointerfaces relevant to ocular applications, where viscosity, stability, adhesion,

controlled release and patient acceptability are critical parameters. The group's work on active and smart functionalities could also inspire new approaches for responsive or multifunctional ocular delivery platforms.

The team could also contribute its strong know-how in micro- and nanoencapsulation, including technologies such as nanospray drying and ultrasonic atomisation. These approaches could be explored for the encapsulation, protection and controlled release of therapeutic molecules, antioxidants, anti-inflammatory compounds, peptides, nucleic acids or other sensitive agents relevant to visual health. This could help address common translational barriers in ophthalmology, such as poor stability, limited bioavailability, short residence time and inefficient delivery across biological barriers.

Another important added value is the group's capacity for advanced morphological and functional characterisation of micro- and nanoparticles, using techniques such as TEM, SEM, CryoTEM, XPS and SAXS. This expertise could support the rational design and validation of nanostructured systems by linking particle morphology, surface properties, internal structure and functional performance.

Finally, the group covers a broad translational range from TRL 2 to TRL 7, which is particularly relevant for a consortium aiming to move from research concepts towards patient benefit. Its interdisciplinary background in nanotechnology, biomaterials, formulation engineering and functional materials could open new routes for ophthalmic innovation, especially in areas such as smart ocular formulations, sustained delivery systems, protective bioactive carriers, diagnostic-enabling materials or sustainable biomaterial platforms.

3. Expertise, assets and collaboration interests

Core expertise, methods and platforms you could contribute
Responsive and smart bioactive delivery systems based in biopolymers *Functional modification of biopolymers properties by self-assembling methodologies* *Organ on a chip microfluidic devices for assessing physiological behaviour* *Advanced micro and nanoencapsulation technologies (nanospraydryer, ultrasonic atomization)* *Morphological characterization of micro and nanoparticles* *Functional characterization of biopolymers*

Infrastructure, equipment, facilities or datasets relevant to consortium work
Electron Microscopy and X-Rays *The Electron Microscopy and X-Rays (EMX) facility is a core multi-user facility that features cutting-edge instrumentation, techniques and expertise required for the characterisation of samples in the physical and life sciences.* *This facility focuses on materials and biological research, the development of novel techniques and instrumentation, as well as providing training, technical support and consultation in the areas of electron microscopy and spectroscopy. The centre facility houses several electron microscopes that can probe the physical, electronic, biological, and chemical structure of matter down to the atomic scale.* *These instruments are coupled with advanced in-situ holders, in which the environment is controlled to match near-realistic conditions of operation, and the sample's behaviour is recorded dynamically in real time. State-of-the-art support facilities also are available, including standard specimen preparation equipment and an image analysis laboratory.* • *Transmission Electron Microscopy (TEM)* • *Cryogenic Transmission Electron Microscopy (Cryo-TEM)* • *Scanning Electron Microscopy (SEM)* • *X-Ray Photoelectron Spectroscopy (XPS)* • *X-Ray Diffraction (XRD)* • *Small-Angle X-Ray Scattering (SAXS)*

- *Sample preparation for electron microscopy*
- *In Situ TEM Holders*
- *Software for imaging analyses*

Material Testing

The Material Testing facility uses complementary techniques to determine the physical and chemical properties of the different types of materials.

Thermogravimetric analysis (TGA) is used to measure changes in the weight of a sample as a function of temperature while the sample is subjected to a controlled temperature program in a controlled atmosphere. It is commonly used to characterize the thermal stability, composition, and decomposition kinetics of materials. Nitrogen adsorption-desorption isotherms at -196 °C are used to characterise the porosity and surface area of materials. ICP-OES (Inductively Coupled Plasma Optical Emission Spectroscopy) is used to determine the elemental composition of materials. UV-accelerated weathering tester is used to simulate the effects of sunlight exposure on materials. These chambers are designed to subject samples to controlled levels of ultraviolet (UV) radiation, temperature, and humidity, replicating the conditions that materials experience when exposed to outdoor environments over extended periods. Salt spray testing is a standardised method used to evaluate the corrosion resistance of materials and coatings.

- Thermogravimetric Analysis / Differential Scanning Calorimetry (TGA/DSC)
- Optical Emission Spectrometry (ICP – OES)
- Nitrogen adsorption-desorption isotherms at -196 ºC
- Taber Abrasion Testing
- Salt Spray and Cyclic Corrosion Testing
- Accelerated Weathering Testing
- Universal Materials Testing
- Rheology of Soft Materials

Core fabrication areas

- (Thin) film deposition and Material Growth
- Etching, Ashing and Micromachining
- Optical and E-beam Lithography
- Advanced Packaging, Annealing and Back-End Processes
- Metrology, Inspection and Wafer-Scale Device Testing

Nanophotonics & Bioimaging

The Nanophotonics & Bioimaging (NBI) facility features cutting-edge optical instrumentation and the know-how required for the characterisation of material and life sciences samples.

Techniques available

Advanced optical microscopes

In-house custom-developed microscopes and spectroscopy solutions

Materials, particles, molecules and cells analysis

Optical spectroscopy solutions

Nonlinear microfabrication

Data analysis for microscopy, spectroscopy and flow cytometry

- Advanced Optical Microscopy
- Optical Spectroscopy
- Nanoparticle, Protein and Cell analysis
- Custom Multi-Modal Microscopy and Microfabrication Solutions

Training capacity and knowledge-transfer potential
Scientific: The technologies to manipulate biopolymers to obtain new functionalities The technologies for micro and nanoencapsulation The methodologies for assessing bioactive delivering

Technical:

The INL Postgraduate School is designed especially for Phd candidates pursuing their degree at INL, to equip them with essential skills beyond the lab, offering tailor-made training to enhance their academic and professional growth. The INL Postgraduate School offers complementary and transversal training designed to enhance specific skills that support researchers in their future careers – whether in academia, industry, or entrepreneurship. These training activities help PhD candidates build a well-rounded professional profile, empowering them with a versatile skill set to succeed across diverse career paths. Throughout their time at INL, PhD candidates receive personalised guidance to support their personal and professional development. This includes a structured series of workshops, masterclasses, and peer discussions tailored to each stage of their PhD journey. In the first year, the focus is on career planning, self-assessment, and navigating the early stages of the PhD. The second year emphasises time management, emotional intelligence, and networking skills. In the final year, candidates are prepared for the job market with targeted training on CV writing, interview techniques, and exploring both academic and non-academic career paths.

Others:

The INL Postgraduate School also offer courses in the following topics:

Communication
Data Management
Leadership
Teamwork
Ethics and Responsible Research
Entrepreneurship
Innovation
Career Management

What your team would hope to gain from participating

Examples: access to methods, models, technologies, patient-oriented validation, translational pathways, industrial insight, clinical exposure, data science, regulatory know-how, joint publications or longer-term strategic collaboration. ***Again, please think broadly and beyond ophthalmology.***

Pastrana Research Group would hope to gain access to a highly interdisciplinary environment where its expertise in biomaterials, micro- and nanoencapsulation, smart functional systems and advanced particle characterization can be translated beyond its current core areas and explored in the context of visual health and patient-oriented innovation. Participation in the consortium would provide the group with valuable exposure to ophthalmology-related needs, clinical challenges and biological models, helping to identify where its enabling technologies could contribute to new delivery platforms, protective biomaterials, responsive formulations or advanced diagnostic-supporting materials. This would be particularly relevant given the group's experience in smart, active and sustainable material systems, micro- and nanostructures, and multifunctional material design.
The team would also benefit from access to translational pathways that are less present in food and biomaterials research, including patient-oriented validation, clinical requirements, regulatory expectations, safety assessment and industrial routes towards medical or health-related applications. Such knowledge would help the group better understand how technologies developed at early or mid TRL levels can be adapted, validated and de-risked for use in more demanding biomedical contexts.
In addition, the consortium would offer opportunities to learn from partners with complementary expertise in ophthalmic models, clinical data, regulatory strategy, device or formulation development, and market-oriented innovation. This could open new research directions for the group, particularly in the adaptation of encapsulation technologies, biomaterial platforms and advanced characterization tools to health-related applications.

Finally, Pastrana Research Group would expect to gain long-term strategic value through joint publications, shared methodologies, cross-sector training, new project ideas and durable collaborations with clinical, academic and industrial partners. Beyond ophthalmology, the consortium could help the group expand its translational capacity and position its nanotechnology-based platforms as versatile enabling technologies for broader health and sustainability challenges.

4. Key people likely to be involved

Please list the key people who may be relevant to the consortium at this stage. *This is not a final, binding or secondment list.*

Name	Position / role	Department / unit	Staff category	Career stage	Email / identifier (optional)
Lorenzo Pastrana	Research group leader		researcher	experienced	Lorenzo.pastrana@inl.int
Mariam Debs	Officer	Communication	managerial	experienced	Mariam.debs@inl.int

Main disciplines represented in your team	biomaterials, encapsulation, bioactive delivery, assessment methodologies, marketing and commercialisation.
Experience in EU / international / multi-partner collaborative projects	Participation in the following projects: Crystal 3, 3DNenonet and coordinate FODIAC
Experience with clinical, translational or patient-facing research (if relevant)	x Yes ☐ No ☐ Not applicable | Short note: Participation in the European project Bugs for urate

5. Evidence of capacity and track record

Up to 5 relevant publications / datasets / software / products / services / other achievements

Please prioritise achievements most relevant to your possible contribution. Please include DOI, URL or identifier.

Pirouni, A.; Drosou, C.; Koskinakis, S.E.; Stergiopoulos, C.; Amado, I.R.; Fuciños, P.; Pastrana, L.; Mishra, P.; Krokida, M. Development of Antimicrobial Textile Coatings Through Encapsulation of ZnO in Electrospun PLA Fibers. *Coatings* 2026, *16*, 672. https://doi.org/10.3390/coatings16060672

Artur J. Martins, Lara Perdigão, Catarina Gonçalves, Isabel R. Amado, Cristiano S. Abreu, António A. Vicente, Rosiane L. Cunha, Lorenzo M. Pastrana, Miguel A. Cerqueira,
Beta-carotene-loaded Oleogels: Morphological analysis, cytotoxicity assessment, in vitro digestion and intestinal permeability, Food Chemistry, Volume 465, Part 2, 2025, 142085, ISSN 0308-8146, https://doi.org/10.1016/j.foodchem.2024.142085.

Marta Vinha Vieira, Shaba Noore, Brijesh Tiwari, Colm O'Donnell, Catarina Gonçalves, Lorenzo M. Pastrana, Pablo Fuciños, Enhancing the stability and functionality of phycobiliproteins as natural food colourants through microparticle formulation, Food Chemistry, Volume 465, Part 2, 2025, 142077, ISSN 0308-8146, https://doi.org/10.1016/j.foodchem.2024.142077

Calero, V., Rodrigues, P.M., Dias, T. et al. A miniaturised semi-dynamic in-vitro model of human digestion. Sci Rep 14, 11923 (2024). https://doi.org/10.1038/s41598-024-54612-w

Ana M. Pinto, Raquel Pereira, Artur J. Martins, Lorenzo M. Pastrana, Miguel A. Cerqueira, Sanna Sillankorva,
Designing an antimicrobial film for wound applications incorporating bacteriophages and ε-poly-l-lysine,
International Journal of Biological Macromolecules, Volume 268, Part 2, 2024, 131963, ISSN 0141-8130,
https://doi.org/10.1016/j.ijbiomac.2024.131963.

Up to 5 relevant previous projects, grants, networks or collaborations

Please provide detailed information (Title, Acronym, funding agency, GA-number, dates). Please indicate your role (coordinator / beneficiary / partner / service provider / collaborator). Indicate if international.

Title: COVICOAT Antiviral edible coating for fresh food
Funded by: EIT Food

Partners: International Iberian Nanotechnology Laboratory and EROSKI
Date: 2020 -2020
Coordinator: Lorenzo Pastrana

Title: TrustEat Building a trusty future food system by using blockchain tech
Funded by: H2020
Partners: International Iberian Nanotechnology Laboratory, WUR and IBM,
Date: 2020 -2023
Coordinator: Lorenzo Pastrana

Title: CRYSTAL3 Commercial & Research Opportunity for Cysteinyl Leukotriene Signalling in Ocular & CNS Dysfunction, Cancer and Cardiovascular Disease
Funded by: H2020
Partners: International Iberian Nanotechnology Laboratory and more
Date: 2021 -2024
(WP): Lorenzo Pastrana

Title: IBEROS+. Instituto de biofabricación en red para el envejecimiento saludable
Funded by: POCTEP
Partners: International Iberian Nanotechnology Laboratory, Universidad de Vigo (coordinador), Universidade do Porto and more
Date: 2023 -2026
(WP): Lorenzo Pastrana

Title: FEAST Fostering European cellular Agriculture for Sustainable Transition Solutions
Funded by: H2020
Partners: International Iberian Nanotechnology Laboratory Universidade de Lisboa, WUR, GFI and more
Date: 2023 -2026
(WP): Lorenzo Pastrana

Patents, protected know-how, clinical protocols, validated tools or other translational assets (if any)
Please indicate only what can be shared at this stage.

6. Staff-exchange capacity (only rough estimation at this stage)

Please provide only a rough, non-binding estimate of your organisation's/team's capacity to send and/or host staff **over the 48-month duration of the project.**

Could your organisation/team second staff?	x Yes ☐ Maybe ☐ No capacity to send staff on secondment
Could your organisation second staff categories beyond researchers (where relevant to R&I activities)?	x Only Researchers ☐ Technical Staff/Innovation x Managerial / project support x Admin Staff ☐ Other profiles
Could your organisation host incoming staff from academic and/or non-academic sectors?	☐ Academic only ☐ Non-academic only x Both ☐ No capacity to host
Administrative / HR / legal considerations that may affect mobility	e.g. visa, insurance, access to facilities, language, employment status, health & safety induction

7. Matchmaking information: Summary for the internal participant database

Capabilities you would most like other interested participants to notice
Please summarise the 2-3 strengths that should be first visible in an internal searchable profile. Encapsulation technologies Biopolymers engineering Organ on a chip devices

Capabilities or expertise you would especially like to access through the consortium
Please indicate the types of teams, methods, infrastructures or translational pathways you would like to connect with in the consortium. Access to cutting-edge infrastructures and specialised technologies Improved translational and methodological standardisation Long-term international collaboration frameworks

Please Confirm:

Open to being contacted by other interested participants through the coordination team's mediation?	x Yes ☐ No
☐ We understand that the information contained in this form will be searchable and accessible to other interested participants within the pre-consortium process.	

8. Final remarks and query items

Please note any uncertainties, prerequisites, internal approvals needed, or items that should be considered further before the consortium is consolidated.
Examples: PIC pending approval, institutional approval, legal status, hosting rules, IP constraints, data access, ethics, clinical permissions, staffing continuity, resource availability. The participation has to be approved by the DG

Internal use and confidentiality statement

The information provided in this form will be used only for internal consortium-building purposes in connection with the preparation of a prospective MSCA Staff Exchanges proposal (2027 call) coordinated from Unidad de Excelencia IOBA / University of Valladolid. It may be used by the coordination team to: (i) assess strategic fit and administrative readiness, (ii) prepare participant-related information required at proposal stage, and (iii) build an internal profile database to facilitate contacts among interested participants. The information will not be shared outside this pre-consortium context without prior permission from the contributing organisation / contact person, except where later required for the formal preparation of a proposal with the explicit involvement of that organisation. Submission of this form confirms that the contributor agrees to this internal use. Organisations may request correction, update or withdrawal of their information at any time by contacting the coordination team.

x Understood

Name: Lorenzo Pastrana
Date completed: 2 /6 / 2026

(Coordination team: ivan.fernandez.bueno@uva.es; yolanda.alvarez.teruel@uva.es)